Body Copilot

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Body Copilot is designed to support insufficiently active adults who want to become more active. The long-term goal is to support health and well-being through increasing physical activity while limiting the effort it takes to plan and reschedule exercise [1]. Open time slots on a calendar may not fit someone's exercise routine or preferences. The system will provide a weekly exercise schedule based on the user's calendar, previous workout history, previous wearable measurements, and preferences. Preferred activities, workout duration, and preparation or travel time will also be taken into account. Users will see their schedule alongside the suggested workout blocks and can decide whether to accept the recommendation or change it depending on how much planning they want to do. Accepted workouts will remain in place unless a conflict arises or the user decides to change them. If a conflict arises, Body Copilot will suggest a new time that fits their schedule.

Existing approaches include activity trackers with feedback, improving adherence through reminders, exercise planning with ML models, and calendar scheduling to help find times to work out. A review of 121 trials found that activity monitors that provide feedback increased physical activity [2]. A review including 41 trials found that planning interventions could help increase activity [3]. HeartSteps provides walking recommendations based on the person's context [4]. Reclaim focuses on scheduling habits around a person's calendar and preferences [5]. PlanFitting is a conversational AI system that provides personalized exercise plans [6]. PRO-Fit is one of the most similar systems found. It combines calendar availability, activity preferences, and exercise scheduling, and uses machine learning to recognize activities [7]. Activity trackers and reminders can support adherence, but these alone cannot resolve scheduling conflicts. Reclaim can help with scheduling, but the article does not describe using past wearable measurements to help choose between workout times [5]. PRO-Fit uses wearable data to recognize activity types, but its study does not test whether physiological measurements improve predictions of future activity start times [7]. We will test whether adding wearable measurements improves activity timing predictions beyond time of day, day of the week, and past activity history. Our proposed advantage and differentiator is using past wearable data to help choose between workout times that already fit the user's schedule and preferences. LifeSnaps dataset includes activity start times and repeated wearable measurements allowing us to test this comparison.

The LifeSnaps dataset includes Fitbit Sense data from 71 participants across 2 study rounds [8]. We will parse the data, remove duplicates, handle missing or invalid measurements, and exclude periods with insufficient data. A logistic regression model will then predict whether a recorded activity will begin within each one-hour window from 6 a.m. to 10 p.m. The model will use past activity history and wearable measurements, including heart rate, resting heart rate, steps, heart rate variability (HRV), and sleep duration depending on the available data. The model will only use previous available data when creating the weekly plan. For each possible workout window, the model will take time of day and the day of the week into account. We will use the recorded activity to train and test these predictions. Calendar availability and required preparation and travel time will be hard constraints, while workout duration and preferences will be flexible within limits set by the user. The model's predictions will help rank the remaining workout windows. If no workout fits, it will say so.

We plan to train the model on the earlier days, check its predictions and settings on the middle days, and test the finished model on the later days from the same participants. We will keep the seven-day predictions from overlapping across training, validation, and testing. We will compare a logistic regression model using time of day, day of the week, and past activity with the same model but with wearable measurements included. We also plan to compare precision, recall, and false-positive rates for both models and compare performance with rules based on time of day and past activity times. Separately, we will test Body Copilot's recommendations using simulated calendars and chosen preferences. We will ensure the system treats calendar commitments as hard constraints, allows enough time for exercise, preparation, and travel, and can suggest alternatives when conflicts arise. We will report these scheduling results separately from the prediction performance.

We identified that missing records and differences between people may limit performance. The Fitbit Sense may automatically record everyday movement rather than planned exercise. We will use earlier records to estimate whether participants meet recommended activity levels while considering missing data and Fitbit's estimates. Predicting when someone would exercise also does not establish the healthiest time for exercise. These simulated schedules and tests cannot tell us whether a real user prefers the recommendations, spends less time planning, or exercises more consistently. We will document the code and evaluation on GitHub so others can follow and reproduce our work.

References

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